King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam II Term 242

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. One statement is False about $f(x) = x^3 + 3x + 1$

(a) There is an interval $[a, b]$, $a < b$, where Rolle's Theorem is applicable

(b) $f(x)$ is differentiable over $(-\infty, \infty)$

(c) $f'(-1) = f'(1)$

(d) $f(x)$ has at most one x-intercept

(e) $f(x)$ has no point where the tangent line is horizontal

2. If $f(x) = e^x \sin x$, then $f'''(0) =$

(a) 2

(b) π

(c) $\frac{1}{2}e^{\pi}$

(d) $\frac{1}{2}$

(e) 1

3. If the range of $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$ over the domain $[-2, 3]$ is $[m, M]$, then $m + M =$

(a) 24

(b) 2

(c) -30

(d) 61

(e) 29

4. Starting with $x_1 = 2$, the second Newton's approximation x_2 to the root of $x^3 - 2x - 5 = 0$ is

(a) 2.01

(b) 1.99

(c) 2.10

(d) 2.00

(e) 1.89

5. If $y = \sin(\cos^{-1} x)$, then $y' =$

(a) $\dfrac{x}{\sqrt{1-x^2}}$

(b) $\dfrac{2}{\sqrt{1-x^2}}$

(c) $\dfrac{-1}{\sqrt{1-x^2}}$

(d) $\dfrac{-x}{\sqrt{1-x^2}}$

(e) $\dfrac{1}{\sqrt{1-x^2}}$

6. If $y = \ln\left(\dfrac{1+e^x}{1-e^x}\right)$, then $y' =$

(a) $\dfrac{1}{2}e^x\left(\dfrac{1+e^x}{1-e^x}\right)$

(b) $\dfrac{e^x}{(1-e^x)^2}$

(c) $\dfrac{2e^x}{1-e^{2x}}$

(d) $\dfrac{1+e^x}{1-e^{2x}}$

(e) $\left(\dfrac{1+e^x}{1-e^x}\right)^2$

7. The critical number of $f(x) = x^{-2}\ln x$ is

(a) e^{-2}

(b) $\dfrac{1}{\sqrt{e}}$

(c) $\sqrt{e}$

(d) $e\sqrt{e}$

(e) $\dfrac{1}{2}e$

8. If $f(x) = \left(2 + (x^2+1)^4\right)^3$, then $f'(1) =$

(a) $3 \cdot 2^9$

(b) $3^2 2^6$

(c) $3^5 2^8$

(d) $3^3 2^7$

(e) $3^4 2^6$

9. If $y = x^{\ln x}$, $x > 0$, then $y'(e) =$

(a) e

(b) 2

(c) e^2

(d) $2e$

(e) $\dfrac{1}{2}$

10. The slope of the tangent line to the curve $y = \dfrac{1 + \csc x}{1 - \csc x}$ at $\left(\dfrac{\pi}{6}, -3\right)$ is equal to:

(a) $2\sqrt{3}$

(b) -3

(c) $\sqrt{3}$

(d) $-4\sqrt{3}$

(e) $\dfrac{2}{3}\sqrt{3}$

11. All edges of a cube are expanding at a rate of 6 centimeters per second. How fast is the surface area changing when each edge is 3 cm?

(a) 256 cm^2/sec

(b) 216 cm^2/sec

(c) 72 cm^2/sec

(d) 144 cm^2/sec

(e) 168 cm^2/sec

12. The sum of all critical numbers of $f(x) = \sin^2 x + \cos x$ over $(0, 2\pi)$ is

(a) $\dfrac{3}{2}\pi$

(b) 3π

(c) 5π

(d) $\dfrac{7}{2}\pi$

(e) $\dfrac{5}{2}\pi$

13. The equation of the tangent line to the curve $\tan(x+y) = x$ at $(0,0)$ is

(a) $y = 1$

(b) $y = x + \dfrac{\pi}{4}$

(c) $y = x - \dfrac{\pi}{4}$

(d) $y = 0$

(e) $y = -1$

14. If c is the number that satisfies the conclusion of the Mean Value Theorem of $f(x) = \dfrac{x}{x+2}$ over $[1,4]$, then

(a) $\dfrac{1}{2} < c < \dfrac{3}{2}$

(b) $\dfrac{5}{2} < c < \dfrac{7}{2}$

(c) $2 < c < 3$

(d) $3 < c < 4$

(e) $1 < c < 2$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$f' = 3x^2 + 3 > 0$, so f is increasing and Rolle's Theorem never applies: that claim is false.
2	**(a)**	$f'' = 2e^x \cos x$, $f''' = 2e^x(\cos x - \sin x)$, so $f'''(0) = 2$.
3	**(b)**	Critical numbers $-1, 0, 2$; values $33, -4, 1, -31, 28$; $m + M = -31 + 33 = 2$.
4	**(c)**	$x_2 = 2 - \frac{f(2)}{f'(2)} = 2 - \frac{-1}{10} = 2.10$.
5	**(d)**	$y' = \cos(\cos^{-1} x) \cdot \frac{-1}{\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}$.
6	**(c)**	$y = \ln(1 + e^x) - \ln(1 - e^x)$, so $y' = \frac{2e^x}{1-e^{2x}}$.
7	**(c)**	$f' = x^{-3}(1 - 2\ln x) = 0$ gives $\ln x = \frac{1}{2}$, $x = \sqrt{e}$.
8	**(c)**	$f'(1) = 3 \cdot 18^2 \cdot 4 \cdot 8 \cdot 2 = 62208 = 3^5 2^8$.
9	**(b)**	$y' = x^{\ln x} \frac{2\ln x}{x}$, so $y'(e) = e \cdot \frac{2}{e} = 2$.
10	**(d)**	$y' = \frac{-2\csc x \cot x}{(1-\csc x)^2}$; at $\pi/6$ it is $-4\sqrt{3}$.
11	**(b)**	$A = 6s^2$, $A' = 12s\, s' = 12 \cdot 3 \cdot 6 = 216$ cm^2/sec.
12	**(b)**	$f' = \sin x(2\cos x - 1) = 0$: $x = \pi, \frac{\pi}{3}, \frac{5\pi}{3}$; sum 3π.
13	**(d)**	$\sec^2(x + y)(1 + y') = 1$ gives $y'(0, 0) = 0$, so $y = 0$.
14	**(c)**	$f'(c) = \frac{2}{(c+2)^2} = \frac{1}{9}$ gives $c = 3\sqrt{2} - 2 \approx 2.24$.